07 Box-Muller Transform

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Last time, we "Changed Variables"

- Given Z a random variable, what is the distribution of $\phi(Z) = W$?

CDF W

- $P(W \le t) = P(\phi(Z) \le t)$
  $= P(Z \le \phi^{-1}(t))$ CDF Z
- $f_W(t) = \frac{d}{dt} P(Z \le \phi^{-1}(t)) = f_Z(\phi^{-1}(t)) \underbrace{\frac{d}{dt}\phi^{-1}(t)}_{\text{Multiply by derivative of the transformation}}$

What happens if we transform $\begin{pmatrix} X \\ Y \end{pmatrix} \to \begin{pmatrix} W \\ Z \end{pmatrix}$?

<u>Proposition</u> Let $U \subset \mathbb{R}^n$ be an open set and $\varphi : X \to Y$, one-to-one, differentiable and with con't partial derivatives and non-zero Jacobian

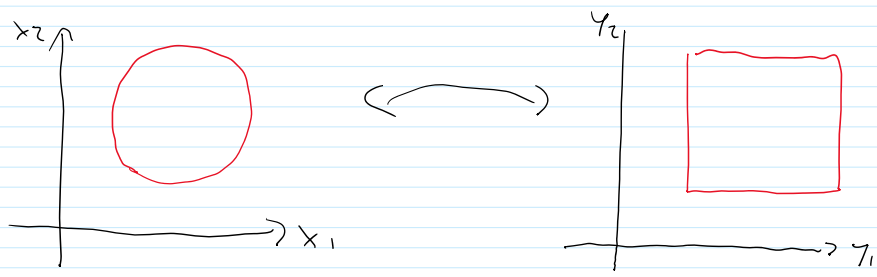
$$J = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & & \\ \frac{\partial y_n}{\partial x_1} & \cdots & \frac{\partial y_n}{\partial x_n} \end{pmatrix}$$

$\hookrightarrow |\det(J)| \neq 0$

Then $\int_{\varphi(U)} f(y)\,dy = \int_U f(x)\,|\det(J)|\,dx$

region region

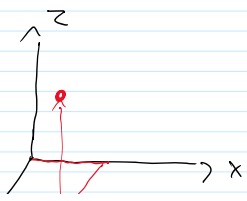
In some sense, a pdf is actually $f(x)\,\boxed{dx}$

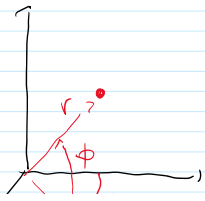


<u>Spherical Coordinates</u> $\mathbb{R}^3$

Cartesian: (x, y, z)

Spherical: (r, θ, ϕ)





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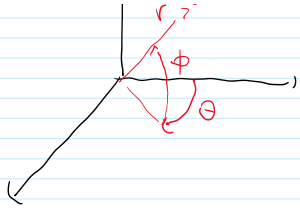

Transformation :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \sin\phi \cos\theta \\ r \sin\phi \sin\theta \\ r \cos\phi \end{pmatrix}$$

$$J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{pmatrix} = \begin{pmatrix} \sin\phi \cos\theta & -r \sin\phi \sin\theta & r \cos\phi \cos\theta \\ \sin\phi \sin\theta & r \sin\phi \cos\theta & r \cos\phi \sin\theta \\ \cos\phi & 0 & -r \sin\phi \end{pmatrix}$$

$$|\det(J)| = r^2 |\sin\phi|$$

Point is that $\iiint_{\varphi(U)} f(x, y, z)\, dx\, dy\, dz$

$$= \iiint_U f(r \sin\phi \cos\theta,\ r \sin\phi \sin\theta,\ r \cos\phi) \times \underbrace{r^2 |\sin\phi|}_{} \, dr\, d\theta\, d\phi$$

plug new variables into $f(\cdots)$

Multiply by Jacobian determinat

Theorem: (Box - Muller)

Let $U \sim Unif[0,1]$, $V \sim Exp(1)$, $U \perp\!\!\!\perp V$ Independent

Then for $X = \sqrt{2V} \cos(2\pi U)$

$Y = \sqrt{2V} \sin(2\pi U)$ Independent

Then $X, Y \sim N(0, 1)$ and $X \perp\!\!\!\perp Y$

Proof: Find the density function for X and Y

- We write U, V in terms of X, Y

$$V = \frac{1}{2}(X^2 + Y^2)$$

$$U = \frac{1}{2\pi} \arctan(Y/X)$$

- Take partial derivatives

$$\frac{\partial v}{\partial x} = x \qquad \frac{\partial v}{\partial y} = y \qquad \checkmark$$

$$\frac{\partial u}{\partial x} = \frac{1}{2\pi} \frac{1}{1+(y/x)^2}\left(\frac{-y}{x^2}\right) = \frac{-y}{4\pi\, v(x,y)}$$

$$\frac{\partial u}{\partial y} = \frac{1}{2\pi} \frac{1}{1+(y/x)^2}\left(\frac{1}{x}\right) = \frac{x}{4\pi\, v(x,y)}$$

what we are solving for

Jacobian

- Do Change of Variables

$$f_{X,Y}(x,y) = f_{U,V}(u,v)\left|\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}\right|$$

$f_U(u)\, f_V(v)$

$$= \mathbb{1}[u(x,y) \in [0,1]]\; e^{-v(x,y)} \left|\frac{-y^2}{4\pi\, v(x,y)} - \frac{x^2}{4\pi\, v(x,y)}\right|$$

Unif[0,1] Exp(1) Jacobian

$$= (1)\, e^{-\frac{1}{2}(x^2+y^2)} \frac{1}{4\pi}\left|\frac{-y^2-x^2}{\frac{1}{2}(x^2+y^2)}\right|$$

$$= \frac{1}{2\pi}\exp\left(-\frac{1}{2}(x^2+y^2)\right) \leftarrow \text{pdf for } N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & \cdot\\ \cdot & 1\end{pmatrix}\right)$$

$\square$